

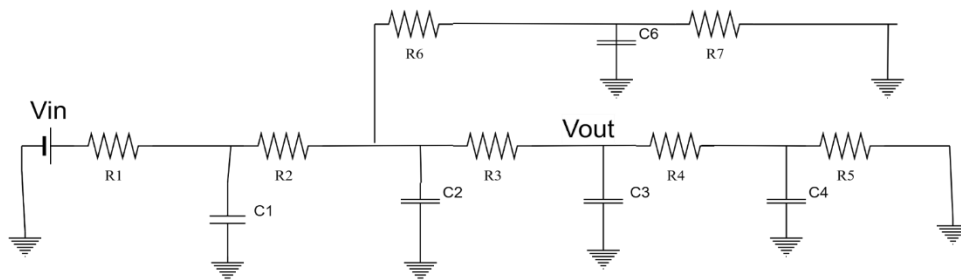
Delay

Q1.

Part1: The **Texas Instrument** company's analog engineers are designing a **CMOS 3- input NAND gate** considering delay and power consumption. They have chosen *GaN* semiconductor which has $\mu_n = 3\mu_p$.

They have **fixed** the width scaling factor $k_n = 2$ for **NMOS** transistors and want to meet **equal fall and rise resistance**. Later, they are informed that the **NAND gate** will drive a total load of **three NOT gates** and **two OR gates**.

Part 2: There is a **RC tree** network given below.



(a)	Find the width scaling factor for PMOS (k_p) to meet the rise and fall resistance specification.	[2]
(b)	Draw the RC equivalent circuit of the NAND gate considering the loads are not connected. Use the Elmore delay model to find the expressions for t_{cdr} , t_{pdr} , t_{cdf} and t_{pdf} .	[7]
(c)	If each NOT gate and OR gate load contributes $3C$, $5C$ unit capacitance respectively, then after connecting the load estimate how many times slower the NAND gate output will operate compared to load disconnected condition considering propagation delay rising for both cases.	[3]
(d)	In Part2 , if the R_1 , R_2 , R_3 , R_4 and R_6 path is on then estimate the worst case rising delay using Elmore delay model associated with V_{in} to be propagated to V_{out} .	[3]

Q-6:

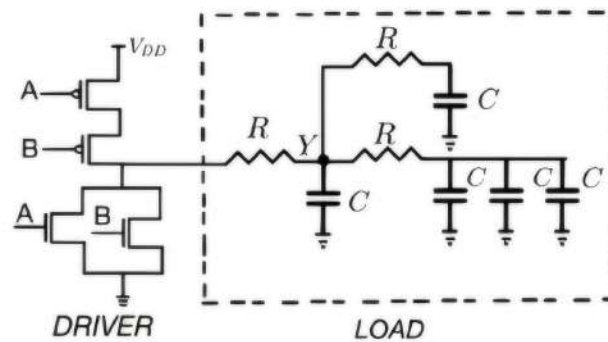


Figure 2: Circuit Diagram for Q.3

(a)	Find the individual transistor widths k_{nMOS} and k_{pMOS} for the <u>Driver</u> to achieve the worst-case effective rise and fall resistance equal to that of a unit inverter, R .	2
(b)	Draw the simplified RC circuit for Figure 2.	5
(c)	i) Derive the expressions for t_{pdr} and t_{pdf} for node Y . ii) Calculate t_{pdr} and t_{pdf} in s (second) from the simplified RC expression from (i) when $R = 1 \text{ k}\Omega$ and $C = 2 \text{ pF}$.	6+2

Q 7.